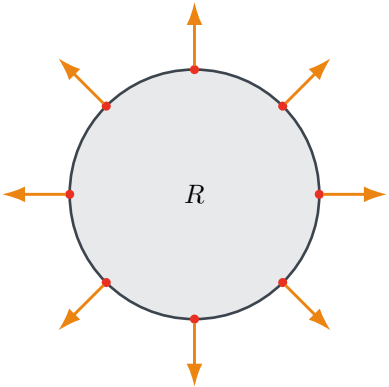
带电导体球的电压力：局部场应取表面两侧场的平均值



电压力沿外法向

$$E_{\text{in}} = 0, \quad E_{\text{out}} = \sigma/\varepsilon_0, \quad E_{\text{自洽}} = \tfrac{1}{2}E_{\text{out}}$$

$$p = \sigma E_{\text{自洽}} = \frac{\sigma^2}{2\varepsilon_0} = \frac{\varepsilon_0 E_{\text{out}}^2}{2}$$

$U = Q^2/(8\pi\varepsilon_0 R)$，固定 Q 作虚位移可复核同一压力。